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UML MODELS

Hotel Automation Software

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Group Members	Suyash Pandey : 2023UG1100 Suraj Kumar : 2023UG1106 Himanshu Raj : 2023UG1092 Satyendra Kumar : 2023UG1105
Institution	Indian Institute of Information Technology Ranchi, Jharkhand

Contents

1 Introduction **3**

2 Use Case Diagram **4**
2.1 Actors 4
2.2 Diagram 4
2.3 Use Case Summaries 4

3 Class Diagram **6**
3.1 Overview 6
3.2 Diagram 6
3.3 Key Relationships 6

4 Sequence Diagrams **7**
4.1 Scenario A — Successful Room Reservation (Walk-In or Advance) 7
4.2 Scenario B — Reservation Rejected (No Suitable Room) 7
4.3 Scenario C — Guest Check-Out with Frequent-Guest Discount 8

5 State-Chart Diagrams **9**
5.1 Room State-Chart 9
5.2 Reservation State-Chart 10
5.3 HotelController State-Chart 10

6 Traceability — UML to SRS **11**

7 Revision History **12**

1 Introduction

This document presents the UML analysis and design models for the Hotel Automation Software, derived from the SRS (v1.0) and the SA/SD document (v1.0). It contains:

- A **use case diagram** capturing the user-observable behaviour.
- A **class diagram** showing the domain and controller classes, their attributes, operations, and relationships.
- **Sequence diagrams** for the three load-bearing runtime scenarios.
- **State-chart diagrams** for the Reservation, Room, and HotelController state machines.

Diagram toolchain

Diagrams are authored in **PlantUML** (source in the `puml/` folder) and rendered to PNG via the PlantUML online editor (`https://www.plantuml.com/plantuml`) or a local `plantuml.jar`. The PNG outputs live in `images/` and are embedded below via `\includegraphics`. See `UML/README.md` for full rendering instructions.

Notation follows UML 2.5.

2 Use Case Diagram

2.1 Actors

The system has four actors corresponding to the user classes defined in SRS §2.3:

- **Hotel Manager** (primary) — configures rooms, tariffs, food menu, and discount policies; reviews occupancy reports.
- **Receptionist** (primary) — manages guest reservations, room allocations, token issuance, and check-out.
- **Catering Services Manager** (primary) — records food and beverage orders against guest token numbers.
- **Guest** (secondary) — the indirect beneficiary; interacts with the system via the Receptionist and Catering Services Manager.

2.2 Diagram

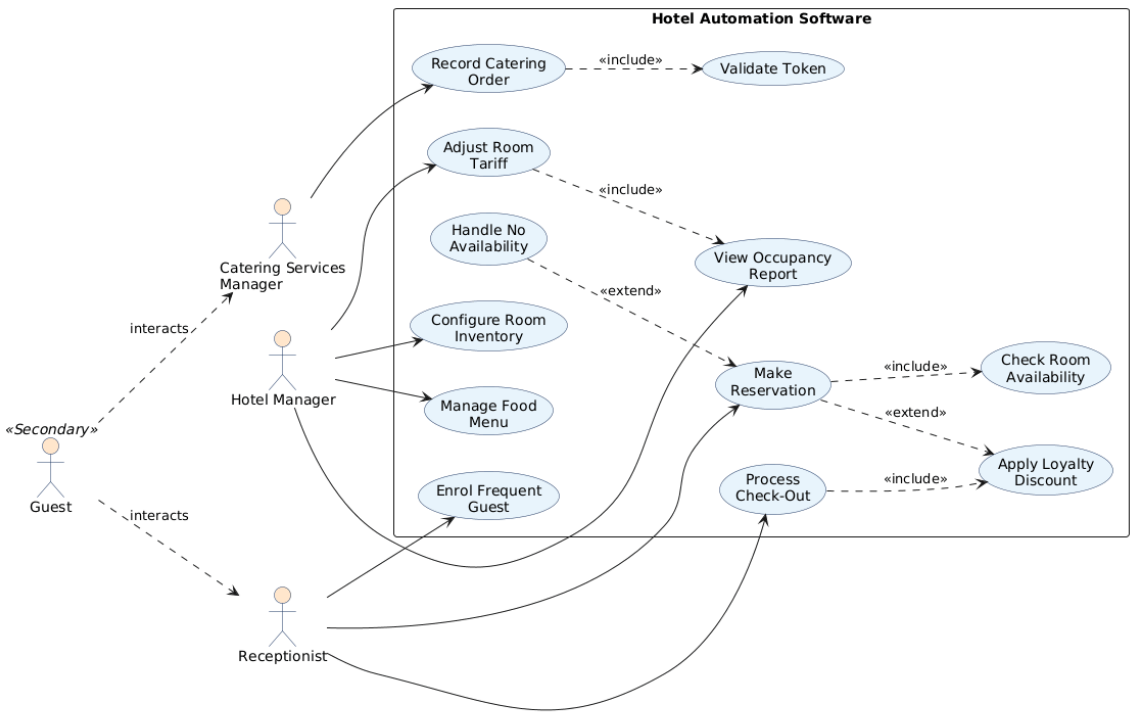

Figure 1: Use Case Diagram. Dashed arrows labelled *<include>* / *<extend>* indicate the corresponding UML relationships.

2.3 Use Case Summaries

Use Case	Summary	SRS Ref
Make Reservation	Receptionist enters guest name, arrival date/time, duration, room type, advance payment, and optional loyalty ID. Includes *Check Room Availability*. Extends with *Apply Loyalty Discount* if identity number is provided.	FR-3.*, FR-4.*
Check Room Availability	Query room inventory for the requested type and date range; return lowest-numbered available room or a NoRoomAvailable flag.	FR-4.1, NFR-R3
Handle No Availability	Extension of Make Reservation when no suitable room exists — compose and display an apology message to the Receptionist.	FR-4.3
Record Catering Order	Catering Services Manager enters token number, food item, quantity, and date/time. Includes *Validate Token*.	FR-5.1–5.4
Process Check-Out	Receptionist enters token number; system computes room and catering charges, applies advance deduction and optional loyalty discount, generates and prints the itemised bill, and marks the room available.	FR-6.1–6.6
Apply Loyalty Discount	Included in Process Check-Out when a frequent-guest identity number is present — look up discount percentage and reduce the total bill accordingly.	FR-7.2, FR-7.3
Enrol Frequent Guest	Receptionist enrols a qualifying guest at check-out; system generates and issues a new loyalty identity number effective from the next stay.	FR-7.1, FR-7.4
Adjust Room Tariff	Hotel Manager selects a room type and specifies a percentage adjustment; system computes the new tariff and logs the change. Includes *View Occupancy Report*.	FR-2.2, FR-2.3
View Occupancy Report	Hotel Manager selects a month; system computes and displays the average occupancy rate.	FR-2.1
Configure Room Inventory	Hotel Manager defines room numbers, bed types, AC types, and base tariffs.	FR-1.1–1.3
Manage Food Menu	Hotel Manager adds or updates food items and their unit prices.	FR-5.3

3 Class Diagram

3.1 Overview

The class model is organised into three logical layers: **UI / Controller** (`HotelController`), **Application core** (`DispatchTransaction`, `ProcessReservation`, `ProcessCheckOut`, `RecordCateringOrder`, `AdjustTariff`, `ViewOccupancy`, `ManageFrequentGuest`, `DBAccessLayer`), and **Domain** (`Room`, `RoomType`, `Tariff`, `Reservation`, `CateringOrder`, `MenuItem`, `FrequentGuest`, `Bill`).

3.2 Diagram

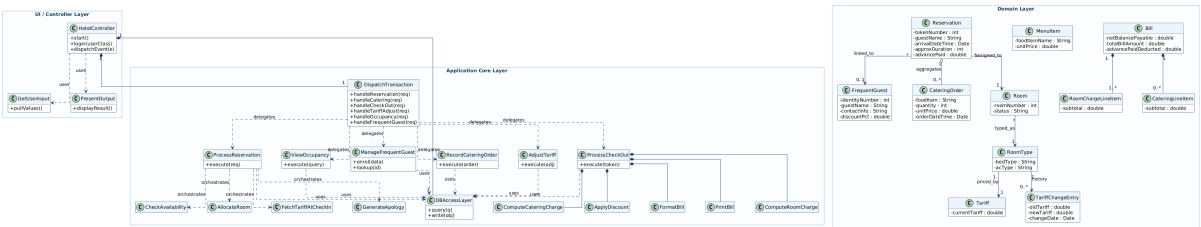

Figure 2: Class Diagram. Filled diamonds: composition; open diamonds: aggregation; open triangles: inheritance; plain arrows: association; dashed arrows: dependency.

3.3 Key Relationships

- `HotelController` composes `DispatchTransaction` and `DBAccessLayer`; it also uses `GetUserInput` and `PresentOutput` for UI mediation.
- `DispatchTransaction` delegates to `ProcessReservation`, `ProcessCheckOut`, `RecordCateringOrde` `AdjustTariff`, `ViewOccupancy`, and `ManageFrequentGuest`.
- `ProcessReservation` orchestrates `CheckAvailability`, `AllocateRoom`, `FetchTariffAtCheckIn`, and `GenerateApology`.
- `ProcessCheckOut` composes `ComputeRoomCharge`, `ComputeCateringCharge`, `ApplyDiscount`, `FormatBill`, and `PrintBill`.
- Every `Room` references one `RoomType`; each `RoomType` references one current `Tariff` and an ordered list of `TariffChangeEntry` objects.
- `Reservation` associates with one `Room` and optionally one `FrequentGuest`; it aggregates zero or more `CateringOrder` objects for billing.
- `Bill` is composed of one or more `RoomChargeLineItem` and zero or more `CateringLineItem` objects.

4 Sequence Diagrams

The three scenarios below exercise the load-bearing paths of the reservation and billing workflows and together cover every transition in the Reservation and Room state machines.

4.1 Scenario A — Successful Room Reservation (Walk-In or Advance)

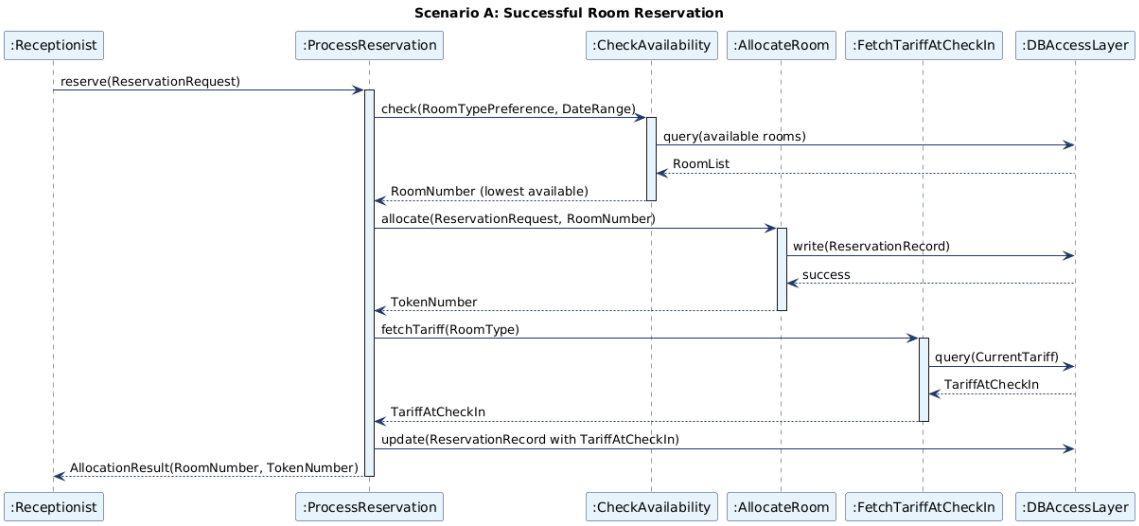

Figure 3: Scenario A. Receptionist submits a reservation request; a matching room is available — system allocates the room, issues a token number, and fetches the tariff at check-in time. (FR-3.1, FR-4.1, FR-4.2, FR-4.4, FR-6.2)

4.2 Scenario B — Reservation Rejected (No Suitable Room)

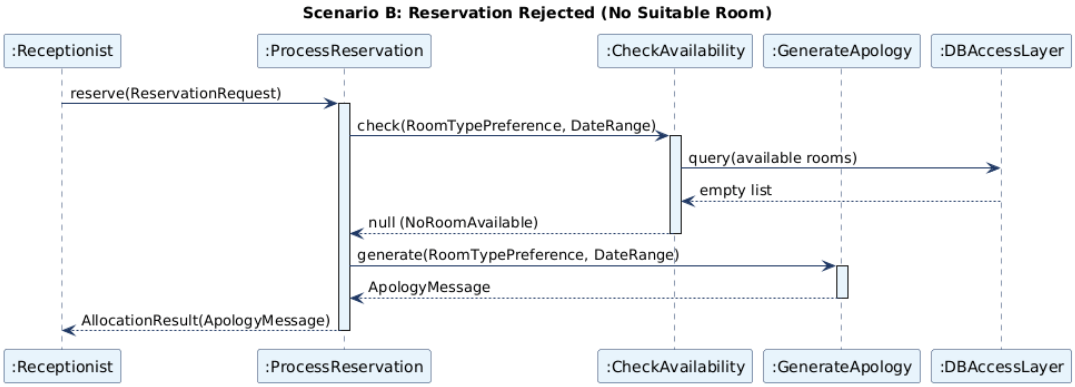

Figure 4: Scenario B. Reservation request is submitted but no room of the requested type is available for the requested dates — system generates and returns an apology message. (FR-4.1, FR-4.3)

4.3 Scenario C — Guest Check-Out with Frequent-Guest Discount

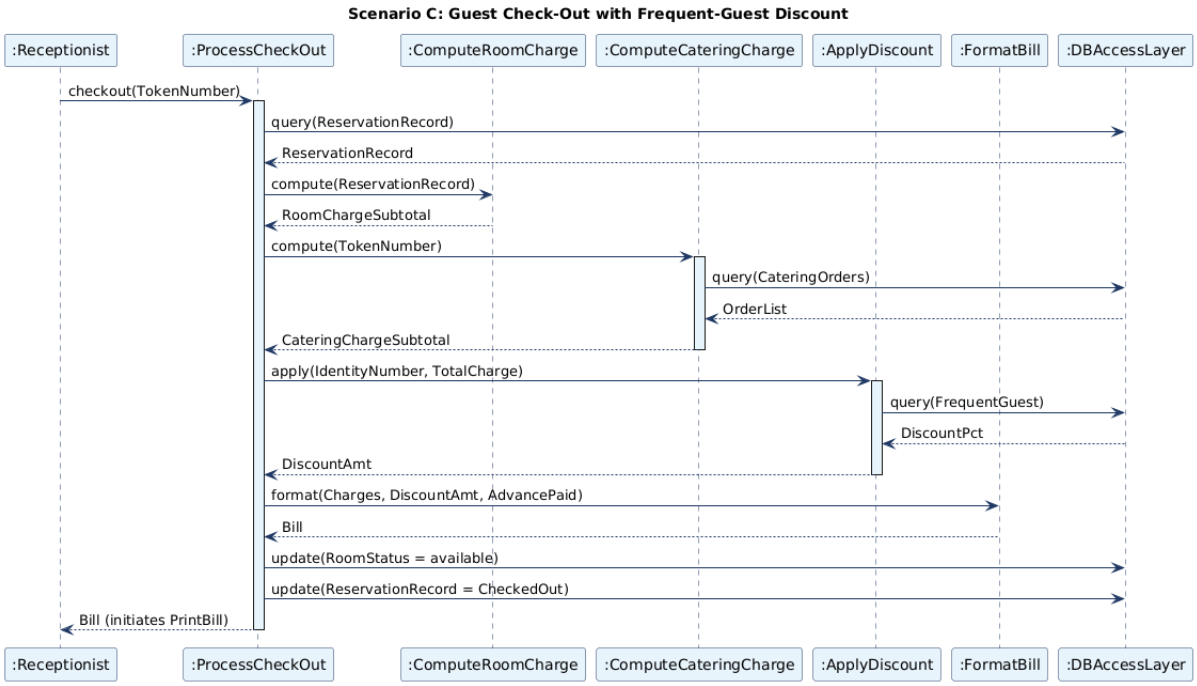

Figure 5: Scenario C. Receptionist initiates check-out; system retrieves reservation and catering records, computes room and catering charges, applies the frequent-guest discount, formats the itemised bill, and marks the room available. (FR-6.1–6.6, FR-7.2–7.3)

5 State-Chart Diagrams

5.1 Room State-Chart

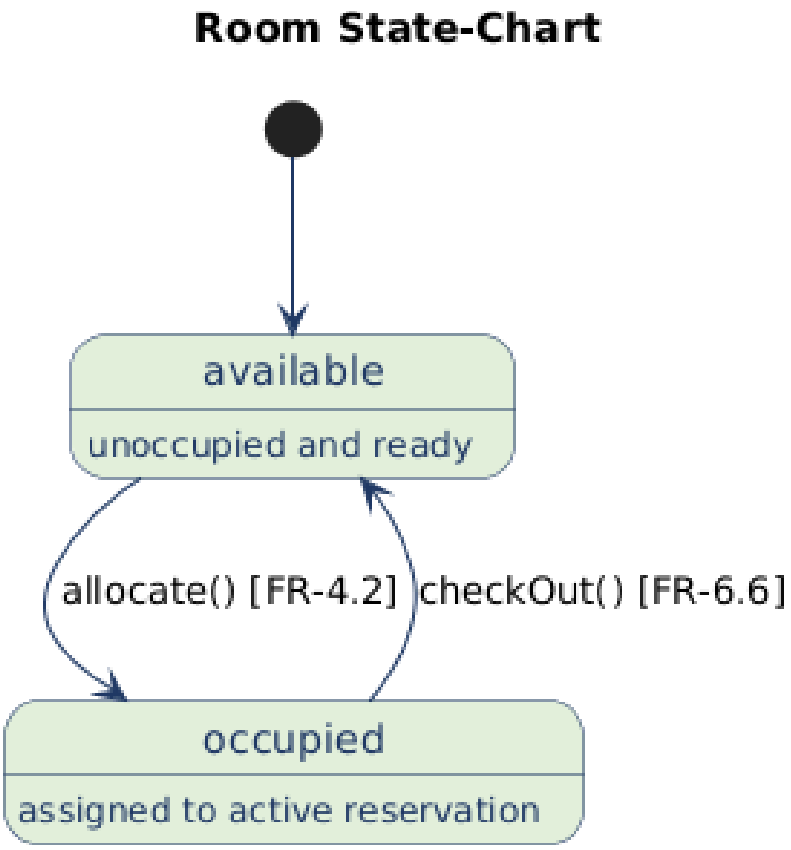

Figure 6: Room state-chart. Two states correspond to FR-1.3's availability partition. Transitions are triggered by successful allocation (FR-4.2) and check-out completion (FR-6.6).

5.2 Reservation State-Chart

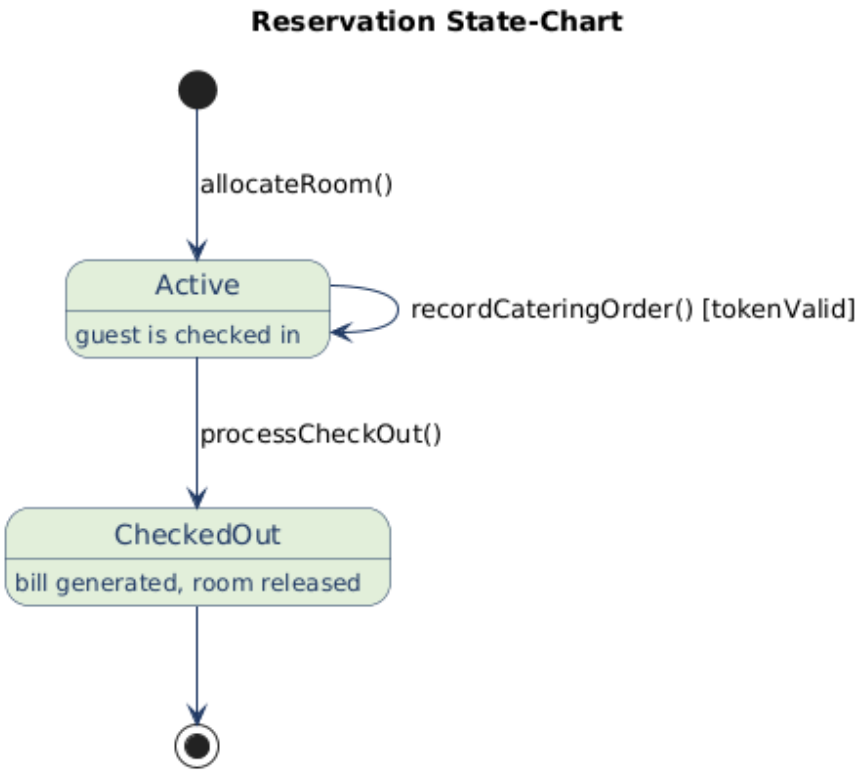

Figure 7: Reservation state-chart. The `[tokenValid]` guard reflects FR-5.2's active-guest check used by the catering module. The `[checkedOut]` transition triggers bill generation and room release. (FR-3.*, FR-4.*, FR-6.1, FR-6.6)

5.3 HotelController State-Chart

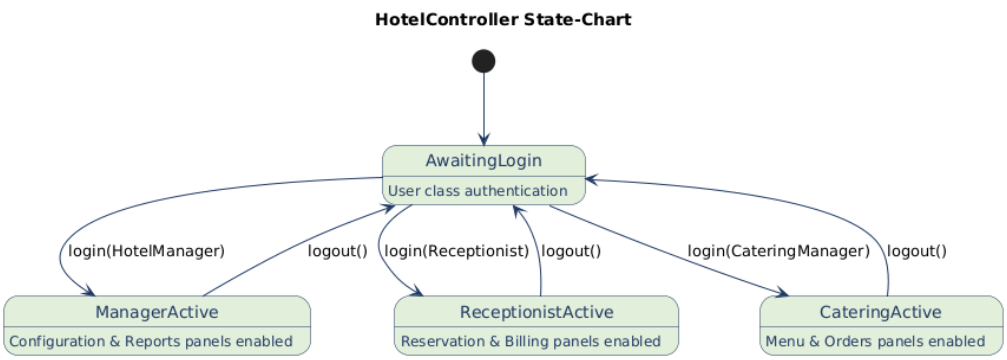

Figure 8: HotelController state-chart covering the GUI lifecycle and role-based panel activation. (FR-8.1, NFR-S1)

6 Traceability — UML to SRS

UML Artefact	Figure	Primary SRS Ref
Use Case: Configure Room Inventory	Fig. 1	FR-1.1–1.3
Use Case: Adjust Room Tariff	Fig. 1	FR-2.2, FR-2.3
Use Case: View Occupancy Report	Fig. 1	FR-2.1
Use Case: Make Reservation	Fig. 1	FR-3.*, FR-4.*
Use Case: Handle No Availability	Fig. 1	FR-4.3
Use Case: Record Catering Order	Fig. 1	FR-5.1–5.4
Use Case: Process Check-Out	Fig. 1	FR-6.1–6.6
Use Case: Apply Loyalty Discount	Fig. 1	FR-7.2, FR-7.3
Use Case: Enrol Frequent Guest	Fig. 1	FR-7.1, FR-7.4
Class: HotelController	Fig. 2	FR-8.1, NFR-S1
Class: ProcessReservation	Fig. 2	FR-3.*, FR-4.*
Class: ProcessCheckOut	Fig. 2	FR-6.1–6.6, FR-7.2–7.3
Class: RecordCateringOrder	Fig. 2	FR-5.1–5.4
Class: ManageFrequentGuest	Fig. 2	FR-7.1–7.4
Class: DBAccessLayer	Fig. 2	NFR-R5
Class: Room, RoomType, Tariff	Fig. 2	FR-1.*, FR-2.*
Class: Reservation, CateringOrder	Fig. 2	FR-3.*, FR-4.*, FR-5.*
Class: Bill	Fig. 2	FR-6.4, NFR-R4
Class: FrequentGuest	Fig. 2	FR-7.1–7.4
Sequence A (reservation → alloc)	Fig. 3	FR-3.1, FR-4.1, FR-4.2, FR-6.2
Sequence B (reservation → apology)	Fig. 4	FR-4.1, FR-4.3
Sequence C (check-out → bill)	Fig. 5	FR-6.1–6.6, FR-7.2, FR-7.3
State-chart: Room	Fig. 6	FR-1.3, FR-4.2, FR-6.6
State-chart: Reservation	Fig. 7	FR-3.*, FR-4.*, FR-5.2, FR-6.1, FR-6.6
State-chart: HotelController	Fig. 8	FR-8.1, NFR-S1

7 Revision History

Version	Date	Author	Description
1.0	April 22, 2026	Project Team	Initial UML models for Hotel Automation Software. Use case diagram (11 use cases, 4 actors), class diagram (three-layer architecture), sequence diagrams for Scenarios A–C, and state-chart diagrams for Room, Reservation, and HotelController. Authored in PlantUML (`puml/`), rendered to PNG, and embedded via `\includegraphics`. All SRS cross-references included.

End of Document — Hotel Automation Software UML v1.0